

(G) from the Glimm–Jaffe–Spencer cluster estimate: explicit derivation of the uniform physical gap

Update (9 August 2026)

Gate R2 is closed (`r2-check.tex`): the GJS norm carries $K_2^{2N(\Delta)}$ and a single K_1 . In Lemma 2.1(ii) and (5.1) below, read $K_1 K_2^{2r}$ for $K_1^{e*} K_2^r$ and $K_1 K_2^{4r}$ for $K_1^{e*} K_2^{2r}$; all conclusions are unchanged.

Abstract

This note executes the (G) side of the weak-coupling closure: from the printed Glimm–Jaffe–Spencer cluster estimate (their Theorem 1.1.7, stated with constants *independent of the cutoff*) we derive

$$\operatorname{spec} H_g \subseteq \{0\} \cup [m_1, \infty) \quad \text{for every } g \in \mathcal{G},$$

with $m_1 > 0$ the (cutoff-independent) decay rate of the cluster estimate, and 0 a simple eigenvalue. All steps are proved: the two-time slab Feynman–Kac transfer, the removal of the $T \rightarrow \infty$ limit by a lower-semicontinuity argument that needs only *second* moments, a density theorem for polynomial cylinder vectors from the factorial moment bound, and the spectral lemma. Three structural simplifications over the sketch in `weak-coupling-strategy.tex` are recorded: (i) the per-vector constants in the decay estimate are allowed to depend on g — only the *rate* must be uniform, so no uniform second moment (M_2) enters the (G) side at all; (ii) no per- g isolation of the ground state is used anywhere (the slab limit $\hat{\psi}_T \rightarrow \Omega_g$ needs only simplicity of the eigenvalue at the bottom); (iii) the fourth-moment truncation of the sketch is replaced by weak lower semicontinuity of the quadratic form $\|e^{-tH_g/2} \cdot\|^2$, which is one-sided in exactly the direction needed. The result is conditional on the printed Theorem 1.1.7 and, for auxiliary second moments, on the class-uniform reading of Theorem 1.1.8 (Gate R1 of `m2-audit.tex`); no other unproved input occurs.

1 Inputs

Notation as in `lppl.tex` and `m2-audit.tex`. In particular $K_g = H_0 + \lambda \int g : \mathcal{P}(\phi_0) :_C dx$ on $\mathcal{H} = L^2(\mathcal{S}'(\mathbb{R}), d\phi_0)$, $E_g = \inf \operatorname{spec} K_g$, $H_g := K_g - E_g \geq 0$, Ω_g the ground state; $d\Phi$ the Euclidean Gaussian with covariance $(-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$, and for a cutoff h in the class $\{0 \leq h \leq 1, \operatorname{supp} h \text{ compact}\}$,

$$dq_h = \frac{e^{-V(h)} d\Phi}{\int e^{-V(h)} d\Phi}, \quad V(h) = \lambda \int : \mathcal{P}(\Phi(z)) : h(z) dz,$$

with the slab cutoffs $h_{T,g}(t, x) = \mathbf{1}_{[-T, T]}(t)g(x)$, $g \in \mathcal{G}$, $T > 0$. Throughout, λ/m_0^2 lies in the GJS weak-coupling region.

The audited source facts used (page references to [1]; see `m2-audit.tex` §2):

Source Fact 1.1 (CE2; printed class-uniform). (p. 594, Thm. 1.1.7.) For localized monomials Q_1, Q_2 of the class (1.1.5) with disjoint localization squares, separated by a strip of width d bounded by two parallel lines (*any orientation*; p. 594),

$$\left| \int Q_1 Q_2 dq_h - \int Q_1 dq_h \int Q_2 dq_h \right| \leq C_7 e^{-m_1 d} \|Q_1\| \|Q_2\|,$$

with $m_1 > 0$ and C_7 independent of h, Q_1, Q_2 . The time-separated situation is native to the source: their Thms. 1.1.10–1.1.11 (p. 596) treat observables “localized at times $> T$ ”.

Source Fact 1.2 (CE1; Gate R1). (p. 595, Thm. 1.1.8.) $|\int Q(x^0) dq_h| \leq C_8 \|Q(x^0)\|$, in the class-uniform reading of Gate R1 of `m2-audit.tex`. Used below only for auxiliary second moments.

Source Fact 1.3 (conventions). (`m2-audit.tex`, Prop. 2.7.) Sharp-time $d\Phi$ -Wick ordering coincides exactly with C -Wick ordering of the time-zero field, $C = (2\mu)^{-1}$; sharp-time functionals of $\Phi(0, \cdot)$ are multiplication operators on $\mathcal{H}$.

2 The observable class and its norm

Definition 2.1. $\mathcal{Q}$ is the set of finite products

$$Q = \prod_{i=1}^r \phi_0(f_i), \quad f_i \in C_c^\infty(\mathbb{R}) \text{ real, } r \in \mathbb{N}_0,$$

($Q = 1$ for $r = 0$), regarded as multiplication operators on $\mathcal{H}$, self-adjoint on their maximal domains. For $s \in \mathbb{R}$ let $Q(s)$ denote the corresponding Euclidean sharp-time monomial $\prod_i \Phi(s, f_i) := \prod_i \int \Phi(s, x) f_i(x) dx$ — a monomial of the GJS class (1.1.5) with r points, all $m_i = 1$, equal times s , and kernel $w = f_1 \otimes \cdots \otimes f_r \in L^2$.

Since all $m_i = 1$, the degree restriction $m_i \leq \kappa$, $\kappa = \deg \mathcal{P}$, of the GJS class is trivially met, and no Wick ordering ambiguity arises: $:\Phi(x): = \Phi(x)$.

Lemma 2.2 (norm facts). *Let $Q \in \mathcal{Q}$ with all $\text{supp } f_i$ contained in an interval of length L , and let $\|Q\|_s$ denote the GJS norm (1.1.9) of $Q(s)$ (sum over the canonical unit-cell localization). Then:*

- (i) (time-shift invariance) $\|Q\|_s = \|Q\|_0 =: \|Q\|_{\text{GJS}}$ for every $s \in \mathbb{R}$;
- (ii) (explicit bound) $\|Q\|_{\text{GJS}} \leq K_1^{e_*} K_2^r r! \prod_{i=1}^r (\sqrt{L+1} \|f_i\|_{L^2})$, with e_* the Gate-R2 exponent;
- (iii) (separation) for $t > 2$, every localized piece of $Q(-t/2)$ is separated from every localized piece of $\tilde{Q}(t/2)$, $\tilde{Q} \in \mathcal{Q}$, by a horizontal strip of width $\geq t - 2$, and the localization squares are disjoint.

Proof. (i) The time shift moves every point’s time coordinate equally, so each point lands in a shifted unit square; the multiset of local degrees $\{N(\Delta)\}$ and the kernel L^2 norms of the localized pieces are unchanged; the sum (1.1.9) has identical terms.

(ii) Decompose $f_i = \sum_j f_i \chi_{[j, j+1)}$; at most $L+1$ terms are nonzero. A localized piece has kernel $\otimes_i f_i \chi_{j_i}$ and norm $K_1^{e_*} \prod_{\Delta} K_2^{N(\Delta)} N(\Delta)! \prod_i \|f_i \chi_{j_i}\|_2 \leq K_1^{e_*} K_2^r r! \prod_i \|f_i \chi_{j_i}\|_2$, using

$\sum_{\Delta} N(\Delta) = r$ and $\prod_{\Delta} N(\Delta)! \leq r!$. Summing over the j_i and applying Cauchy–Schwarz per factor, $\sum_j \|f_i \chi_j\|_2 \leq \sqrt{L+1} \|f_i\|_2$.

(iii) A point at time $-t/2$ lies in a lattice square whose time interval is contained in $[-t/2 - 1, -t/2 + 1]$; a point at time $t/2$, in one contained in $[t/2 - 1, t/2 + 1]$. For $t > 2$ the open horizontal strip $\{-t/2 + 1 < \text{time} < t/2 - 1\}$ of width $t - 2$ separates all squares of the first family from all of the second, and the families are disjoint. $\square$

Corollary 2.3 (two-time cluster bound). *For $Q, \tilde{Q} \in \mathcal{Q}$, $t > 2$, and every cutoff h in the class,*

$$\left| \int Q(-t/2) \tilde{Q}(t/2) dq_h - \int Q(-t/2) dq_h \int \tilde{Q}(t/2) dq_h \right| \leq C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}} \|\tilde{Q}\|_{\text{GJS}}.$$

Proof. Write $Q(-t/2) = \sum_a Q_a$, $\tilde{Q}(t/2) = \sum_b \tilde{Q}_b$ in localized pieces (finite sums). The connected expectation is bilinear, so it equals $\sum_{a,b}$ of connected pieces; each pair is separated by the strip of Lemma 2.2(iii), so Source Fact 1.1 applies with $d = t - 2$; summing, $\sum_a \|Q_a\| = \|Q\|_{\text{GJS}}$ by (1.1.9) and Lemma 2.2(i). $\square$

3 Slab transfer

Throughout this section $g \in \mathcal{G}$ is *fixed*; no uniformity in g is claimed or needed. Set

$$\psi_s := e^{-sH_g} \Omega_0 \quad (s \geq 0), \quad \hat{\psi}_s := \psi_s / \|\psi_s\|, \quad c_g := \langle \Omega_g, \Omega_0 \rangle = \int \Omega_g d\phi_0 > 0,$$

positivity of c_g because $\Omega_g > 0$ a.e. in Q -space and $\Omega_0 \equiv 1$.

Lemma 3.1 (slab FKN, one and two times). *Let $T > 0$, $t \in [0, 2T]$, and let F_1, F_2 be nonnegative measurable functions of the time-zero field. Then*

$$\int F_1(\Phi(-\tfrac{t}{2}, \cdot)) F_2(\Phi(\tfrac{t}{2}, \cdot)) dq_{h_{T,g}} = \frac{\langle F_1 \psi_{T-t/2}, e^{-tH_g} F_2 \psi_{T-t/2} \rangle}{\|\psi_T\|^2}, \quad (1)$$

$$\int F_1(\Phi(s, \cdot)) dq_{h_{T,g}} = \frac{\langle \psi_{T-s}, F_1 \psi_{T+s} \rangle}{\|\psi_T\|^2} \quad (|s| \leq T), \quad (2)$$

as identities in $[0, \infty]$. Both extend to $F_i = Q_i \in \mathcal{Q}$ (and finite real linear combinations) whenever the right-hand sides are finite, by the truncation argument of Lemma 3.3(iii).

Proof. For bounded F_i this is the Feynman–Kac–Nelson formula for the measure $Z^{-1} e^{-V(h_{T,g})} d\Phi$: the Markov property of $d\Phi$ with respect to constant-time lines gives, for $-T \leq t_1 \leq t_2 \leq T$,

$$\int F_1(\Phi(t_1, \cdot)) F_2(\Phi(t_2, \cdot)) e^{-V(h_{T,g})} d\Phi = \langle \Omega_0, e^{-(T+t_1)K_g} F_1 e^{-(t_2-t_1)K_g} F_2 e^{-(T-t_2)K_g} \Omega_0 \rangle,$$

(Nelson; [3, Thm. III.12]; [2]); dividing by the same expression with $F_i = 1$ and cancelling the factors e^{-2TE_g} gives (1)–(2) with $t_1 = -t/2$, $t_2 = t/2$, resp. a single time s , after using self-adjointness of e^{-aK_g} to move semigroup factors to the left slot. Monotone convergence extends both identities to nonnegative measurable F_i , in $[0, \infty]$. $\square$

Lemma 3.2 (convergence of the slab vacuum; no isolation used). *$\hat{\psi}_s \rightarrow \Omega_g$ in norm and $\|\psi_s\| \downarrow c_g > 0$ as $s \rightarrow \infty$. Moreover $\|\psi_{s'}\|/\|\psi_s\| \in [1, \|\Omega_0\|/c_g]$ for all $0 \leq s' \leq s$, and $\rightarrow 1$ when $s, s' \rightarrow \infty$.*

Proof. 0 is a simple eigenvalue of $H_g \geq 0$ with eigenvector Ω_g (GJS p. 588 for measurable g ; GJ II for smooth g). Let $P_0 = |\Omega_g\rangle\langle\Omega_g|$. Then $\psi_s = c_g\Omega_g + e^{-sH_g}(1 - P_0)\Omega_0$ and, by the spectral theorem and dominated convergence, $\|e^{-sH_g}(1 - P_0)\Omega_0\|^2 = \int_{(0,\infty)} e^{-2sE} d\mu_{(1-P_0)\Omega_0}(E) \rightarrow 0$: the spectral measure of $(1 - P_0)\Omega_0$ has no atom at 0 by *simplicity alone*; no gap is invoked. Monotonicity of $s \mapsto \|\psi_s\|$ is $\|e^{-sH}\| \leq 1$; the limit is c_g ; the ratio bounds follow with $\|\Omega_0\| = 1$. $\square$

Lemma 3.3 (second moments and domains). *Let $Q \in \mathcal{Q}$. Then, with C_8 as in Source Fact 1.2:*

- (i) Q^2 is (a finite sum of) monomials of the class (1.1.5) at one time, and $\|Q^2\|_{\text{GJS}} \leq K_1^{e*} K_2^{2r}(2r)! \prod_i (L+1) \|f_i\|_2^2 =: \mathbf{n}(Q)^2$;
- (ii) $\sup_{T>0} \|Q\hat{\psi}_T\|^2 = \sup_{T>0} \int Q(0)^2 dq_{h_{T,g}} \leq C_8 \mathbf{n}(Q)^2$; in particular $\psi_T \in D(Q)$ for every T , and $\Omega_g \in D(Q)$ with $\|Q\Omega_g\|^2 \leq C_8 \mathbf{n}(Q)^2$;
- (iii) for every $a \geq 0$ and $M > 0$, with $Q^{(M)} := Q\mathbf{1}_{\{|Q| \leq M\}}$: $\|(Q - Q^{(M)})\psi_a\| \rightarrow 0$ as $M \rightarrow \infty$.

Proof. (i) is Lemma 2.2(ii) applied to the $2r$ factors of Q^2 , all at one time. (ii): the first equality is (1) with $t = 0$, $F_1 F_2 = Q^2 \geq 0$ (monotone-class version); the bound is Source Fact 1.2 plus (i), uniformly in T since $h_{T,g}$ stays in the class. The statement for Ω_g follows as in `m2-audit.tex`, Lemma 4.3: Q is a closed operator (real multiplication), its graph is weakly closed, $\hat{\psi}_T \rightarrow \Omega_g$ by Lemma 3.2, and $\sup_T \|Q\hat{\psi}_T\| < \infty$. (iii): $\|(Q - Q^{(M)})\psi_a\|^2 = \int |Q|^2 \mathbf{1}_{\{|Q| > M\}} \psi_a^2 d\phi_0 \rightarrow 0$ by dominated convergence, the dominator $|Q|^2 \psi_a^2$ being integrable by (ii). $\square$

Lemma 3.4 (one-point limit). *For $Q \in \mathcal{Q}$ and fixed $t \geq 0$: $\lim_{T \rightarrow \infty} \int Q(\pm t/2) dq_{h_{T,g}} = \langle \Omega_g, Q\Omega_g \rangle =: \omega_g(Q)$.*

Proof. By (2) (extended to Q via Lemma 3.3(iii) and dominated convergence, the dominator $|Q|(\psi_{T-s}^2 + \psi_{T+s}^2)$ being integrable by Lemma 3.3(ii) and Cauchy–Schwarz),

$$\int Q(s) dq_{h_{T,g}} = \left\langle \hat{\psi}_{T-s}, \frac{Q\psi_{T+s}}{\|\psi_T\|} \right\rangle \cdot \frac{\|\psi_{T-s}\|}{\|\psi_T\|}.$$

As $T \rightarrow \infty$: $\hat{\psi}_{T-s} \rightarrow \Omega_g$ strongly; $v_T := Q\psi_{T+s}/\|\psi_T\|$ satisfies $\|v_T\| = \|Q\hat{\psi}_{T+s}\| (\|\psi_{T+s}\|/\|\psi_T\|) \leq C_8^{1/2} \mathbf{n}(Q)$ and $v_T \rightarrow Q\Omega_g$ weakly (bounded, and any weak limit point is $Q\Omega_g$ by weak closedness of the graph of Q , since $\psi_{T+s}/\|\psi_T\| \rightarrow \Omega_g$ by Lemma 3.2); a strongly convergent paired with a weakly convergent sequence gives $\langle \hat{\psi}_{T-s}, v_T \rangle \rightarrow \langle \Omega_g, Q\Omega_g \rangle$; and $\|\psi_{T-s}\|/\|\psi_T\| \rightarrow 1$. $\square$

Lemma 3.5 (two-point upper bound by lower semicontinuity). *For $Q \in \mathcal{Q}$ (real) and $t \geq 0$:*

$$\langle Q\Omega_g, e^{-tH_g} Q\Omega_g \rangle \leq \liminf_{T \rightarrow \infty} \int Q(-t/2) Q(t/2) dq_{h_{T,g}}.$$

Proof. By (1) (extended to Q by Lemma 3.3(iii)): truncate both insertions, use $|Q^{(M)}(-\frac{t}{2})Q^{(M)}(\frac{t}{2})| \leq \frac{1}{2}(Q(-\frac{t}{2})^2 + Q(\frac{t}{2})^2)$ as an integrable dominator on the dq side and norm convergence $Q^{(M)}\psi_{T-t/2} \rightarrow Q\psi_{T-t/2}$ on the Hilbert side),

$$\int Q(-t/2) Q(t/2) dq_{h_{T,g}} = \|e^{-tH_g/2} u_T\|^2, \quad u_T := \frac{Q\psi_{T-t/2}}{\|\psi_T\|}.$$

By Lemmas 3.2 and 3.3(ii), $\|u_T\| \leq C_8^{1/2} \mathbf{n}(Q) \cdot (\|\psi_{T-t/2}\|/\|\psi_T\|)$ is bounded in $T \geq t$, and $u_T \rightharpoonup Q\Omega_g$ (weak closedness of the graph again). Hence $e^{-tH_g/2}u_T \rightharpoonup e^{-tH_g/2}Q\Omega_g$, and weak lower semicontinuity of the norm gives $\|e^{-tH_g/2}Q\Omega_g\|^2 \leq \liminf_T \|e^{-tH_g/2}u_T\|^2$. $\square$

Remark 3.6. Lemma 3.5 is the replacement for the fourth-moment truncation step proposed in `weak-coupling-strategy.tex`: the quadratic form is bounded *from above* in the limit by the slab quantities, which is the only direction the gap proof uses, and it costs only second moments. Equality in Lemma 3.5 is true but not needed.

4 The decay estimate

Proposition 4.1 (connected decay in real time). *Let $Q \in \mathcal{Q}$ be real, $\psi_Q := Q\Omega_g - \omega_g(Q)\Omega_g$. Then for all $t \geq 0$,*

$$\langle \psi_Q, e^{-tH_g} \psi_Q \rangle \leq \widehat{C}_Q e^{-m_1 t}, \quad \widehat{C}_Q := \max\left(C_7 e^{2m_1} \|Q\|_{\text{GJS}}^2, e^{3m_1} \|\psi_Q\|^2\right), \quad (3)$$

where m_1 and C_7 are the (g - and Q -independent) constants of Source Fact 1.1, and $\|\psi_Q\|^2 \leq C_8 \mathbf{n}(Q)^2$ is finite (possibly g -dependent bounds are harmless).

Proof. $\psi_Q \perp \Omega_g$ by construction. Since Q is a real multiplication operator and $e^{-tH_g}\Omega_g = \Omega_g$,

$$\langle \psi_Q, e^{-tH_g} \psi_Q \rangle = \langle Q\Omega_g, e^{-tH_g} Q\Omega_g \rangle - \omega_g(Q)^2.$$

Let $t > 2$. By Lemma 3.5, then Corollary 2.3 (with $h = h_{T,g}$, constants independent of T and g), then Lemma 3.4,

$$\begin{aligned} \langle Q\Omega_g, e^{-tH_g} Q\Omega_g \rangle &\leq \liminf_{T \rightarrow \infty} \int Q(-\tfrac{t}{2}) Q(\tfrac{t}{2}) dq_{h_{T,g}} \\ &\leq C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}}^2 + \lim_{T \rightarrow \infty} \int Q(-\tfrac{t}{2}) dq_{h_{T,g}} \int Q(\tfrac{t}{2}) dq_{h_{T,g}} \\ &= C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}}^2 + \omega_g(Q)^2, \end{aligned}$$

using $\liminf(a_T + b_T) \leq \sup_T a_T + \lim_T b_T$ for the convergent disconnected part. This proves (3) for $t > 2$ with constant $C_7 e^{2m_1} \|Q\|_{\text{GJS}}^2$. For $0 \leq t \leq 3$, $t \mapsto \langle \psi_Q, e^{-tH_g} \psi_Q \rangle = \|e^{-tH_g/2} \psi_Q\|^2$ is nonincreasing and bounded by $\|\psi_Q\|^2 \leq e^{3m_1} \|\psi_Q\|^2 e^{-m_1 t}$. Taking the maximum of the two constants covers all $t \geq 0$. $\square$

5 Density of polynomial cylinder vectors

Let $d\mu_g = \Omega_g^2 d\phi_0$, so $\langle Q\Omega_g, Q'\Omega_g \rangle = \mathbb{E}_{\mu_g}[QQ']$ for $Q, Q' \in \mathcal{Q}$.

Lemma 5.1 (factorial moments $\Rightarrow$ exponential moments). *Let $f \in C_c^\infty(\mathbb{R})$ be real with $\text{supp } f$ in an interval of length L . Then for all $n \in \mathbb{N}$,*

$$\mathbb{E}_{\mu_g}[\phi_0(f)^{2n}] \leq C_8 K_1^{e_*(2n)} (K_2 \sqrt{L+1} \|f\|_2)^{2n} (2n)!,$$

and consequently there is $s_0 = s_0(f) > 0$ (independent of g) with $\mathbb{E}_{\mu_g}[e^{s|\phi_0(f)|}] < \infty$ for $0 \leq s < s_0$.

Proof. Apply Lemma 3.3(ii) to $Q = \phi_0(f)^n$ (an element of $\mathcal{Q}$ with $r = n$ equal factors): $\mathbb{E}_{\mu_g}[\phi_0(f)^{2n}] = \|Q\Omega_g\|^2 \leq C_8 \mathbf{n}(Q)^2$ with $\mathbf{n}(Q)^2 \leq K_1^{e*} K_2^{2n} (2n)! ((L+1)\|f\|_2^2)^n$ by Lemma 3.3(i). Under every Gate-R2 reading, $K_1^{e*} \leq K_1^{\max(2,2n)}$, so the displayed bound holds with a geometric constant $K' := K_1 K_2 \sqrt{L+1} \|f\|_2$ (absorbing K_1). Then

$$\mathbb{E}_{\mu_g}[\cosh(s\phi_0(f))] = \sum_{n \geq 0} \frac{s^{2n}}{(2n)!} \mathbb{E}_{\mu_g}[\phi_0(f)^{2n}] \leq C_8 \sum_{n \geq 0} (sK')^{2n} < \infty \quad \text{for } s < 1/K',$$

and $e^{s|x|} \leq 2 \cosh(sx)$. □

Theorem 5.2 (density). $\text{span}_{\mathbb{C}}\{Q\Omega_g : Q \in \mathcal{Q}\}$ is dense in $\mathcal{H}$.

Proof. Since $\Omega_g > 0$ a.e., $\mathcal{H} \ni \psi \mapsto \psi/\Omega_g$ is unitary from $\mathcal{H}$ onto $L^2(d\mu_g)$, and the claim is equivalent to density of the polynomials in the variables $\{\phi_0(f) : f \in C_c^\infty \text{ real}\}$ in $L^2(d\mu_g)$.

Fix a countable family $(e_j)_{j \geq 1} \subset C_c^\infty(\mathbb{R})$, real, dense in $L^2(\mathbb{R})$, and let $\Sigma_k := \sigma(\phi_0(e_1), \dots, \phi_0(e_k))$.

Step 1: orthogonality kills conditional expectations. Let $\psi \in L^2(d\mu_g)$ be orthogonal to all polynomials in $\phi_0(e_1), \dots, \phi_0(e_k)$. Set, for $z = (z_1, \dots, z_k) \in \mathbb{C}^k$,

$$F(z) := \mathbb{E}_{\mu_g} \left[\bar{\psi} \exp \left(\sum_{j=1}^k z_j \phi_0(e_j) \right) \right].$$

By Lemma 5.1 and Cauchy–Schwarz, $\mathbb{E}_{\mu_g}[|\bar{\psi}| e^{s \sum_j |\phi_0(e_j)|}] \leq \|\psi\|_{L^2(\mu_g)} \mathbb{E}_{\mu_g}[e^{2s \sum_j |\phi_0(e_j)|}]^{1/2} < \infty$ for $s < s_1(k)$ (Hölder splits the sum), so F is defined and, by dominated convergence and Morera, analytic in each variable on the polystrip $\{|\text{Re } z_j| < s_1/k\}$, jointly continuous, hence analytic there. All derivatives at $z = 0$ are $\mathbb{E}[\bar{\psi} \prod \phi_0(e_j)^{\alpha_j}] = 0$. By iterated application of the one-variable identity theorem along coordinate directions, $F \equiv 0$ on the polystrip; in particular on $(i\mathbb{R})^k$:

$$\mathbb{E}_{\mu_g} \left[\bar{\psi} e^{i \sum t_j \phi_0(e_j)} \right] = 0 \quad \forall t \in \mathbb{R}^k.$$

The finite complex measure $\rho(B) := \mathbb{E}_{\mu_g}[\bar{\psi} \mathbf{1}_B(\phi_0(e_1), \dots, \phi_0(e_k))]$ on $\mathbb{R}^k$ has vanishing Fourier transform, hence $\rho = 0$, hence $\mathbb{E}_{\mu_g}[\bar{\psi} G] = 0$ for every bounded Σ_k -measurable G ; that is, $\mathbb{E}_{\mu_g}[\psi | \Sigma_k] = 0$.

Step 2: martingale exhaustion. The variance of $\phi_0(f) - \phi_0(e)$ under $d\phi_0$ is $\langle f - e, C(f - e) \rangle \leq \|C\| \|f - e\|_2^2$, and $\mu_g \ll \phi_0$; hence for every real $f \in C_c^\infty$, $\phi_0(f)$ is measurable, modulo μ_g -null sets, with respect to $\bigvee_k \Sigma_k$. Since the $\phi_0(f)$ generate the Borel σ -algebra of $\mathcal{S}'(\mathbb{R})$ (cylinder sets), $\bigvee_k \Sigma_k$ is the full σ -algebra mod μ_g -null sets. If now $\psi \perp$ all polynomials in all variables, Step 1 gives $\mathbb{E}_{\mu_g}[\psi | \Sigma_k] = 0$ for every k , and the martingale convergence theorem gives $\psi = \lim_k \mathbb{E}_{\mu_g}[\psi | \Sigma_k] = 0$ in $L^2(d\mu_g)$. □

6 The spectral lemma and the theorem

Lemma 6.1 (spectral lemma). *Let $H \geq 0$ be self-adjoint on $\mathcal{H}$ with $H\Omega = 0$, $\|\Omega\| = 1$. Suppose $D \subset \mathcal{H}$ is a set of vectors with $\langle \Omega, \psi \rangle = 0$ and $\langle \psi, e^{-tH} \psi \rangle \leq C_\psi e^{-\gamma t}$ ($t \geq 0$) for each $\psi \in D$, and that $\text{span}(D \cup \{\Omega\})$ is dense. Then*

$$\text{spec } H \subseteq \{0\} \cup [\gamma, \infty), \quad 0 \text{ a simple eigenvalue.}$$

Proof. Let $\beta < \gamma$ and $P := P_{[0, \beta]}(H)$. For $\psi \in D$, with μ_ψ the spectral measure of ψ ,

$$\|P\psi\|^2 = \mu_\psi([0, \beta]) \leq e^{\beta t} \int_{[0, \beta]} e^{-tE} d\mu_\psi(E) \leq e^{\beta t} C_\psi e^{-\gamma t} \xrightarrow[t \rightarrow \infty]{} 0,$$

so $P\psi = 0$; also $P\Omega = \Omega$. Hence P maps the dense subspace $\text{span}(D \cup \{\Omega\})$ into $\mathbb{C}\Omega$; by continuity and closedness of $\mathbb{C}\Omega$, $\text{ran } P \subseteq \mathbb{C}\Omega$, and since $P\Omega = \Omega$, $P = |\Omega\rangle\langle\Omega|$. Thus $\text{spec } H \cap [0, \beta] = \{0\}$, simple, for every $\beta < \gamma$. $\square$

Theorem 6.2 ((G), conditional). *Assume the printed Theorem 1.1.7 of [1] (Source Fact 1.1) and the class-uniform reading of Theorem 1.1.8 (Gate R1). Then, in the GJS weak-coupling region, for every $g \in \mathcal{G}$,*

$$\text{spec } H_g \subseteq \{0\} \cup [m_1, \infty),$$

with $m_1 > 0$ the cluster rate, independent of g , and 0 a simple eigenvalue of H_g .

Proof. Fix g . Let $D := \{\psi_Q = Q\Omega_g - \omega_g(Q)\Omega_g : Q \in \mathcal{Q} \text{ real}\}$. Each ψ_Q satisfies the hypothesis of Lemma 6.1 with $\gamma = m_1$ and $C_\psi = \hat{C}_Q < \infty$, by Proposition 4.1. Since $Q\Omega_g = \psi_Q + \omega_g(Q)\Omega_g$ and complex spans of real Q 's exhaust $\mathcal{Q}$, $\text{span}(D \cup \{\Omega_g\}) \supseteq \text{span}_{\mathbb{C}}\{Q\Omega_g : Q \in \mathcal{Q}\}$, which is dense by Theorem 5.2. Lemma 6.1 applies. $\square$

Remark 6.3 (where uniformity does and does not enter). Only the *rate* m_1 and the cluster constant C_7 are g -uniform, and these come printed in Source Fact 1.1. The per-vector constants $\hat{C}_Q$ contain $\|\psi_Q\|^2$, which the proof bounds via Gate R1 uniformly in g — but g -dependence here would be harmless: Lemma 6.1 never sums over ψ . Consequently (M_2) is *not an input to (G)*; the two hypotheses of `lpp1.tex` are logically independent downstream of the same source estimates. This corrects the impression left by the execution order of `weak-coupling-strategy.tex` that the moment bound had to be discharged first.

Remark 6.4 (no circularity). Nothing in §§3–6 assumes any gap: Lemma 3.2 uses only simplicity of the lowest eigenvalue (an audited statement of the source for measurable g), and all limits are taken at fixed g with the gap appearing only in the *conclusion*. In particular the argument does not use, and does not need, per- g isolation of E_g as an a priori input — isolation, with the uniform bound m_1 , is part of the output.

Remark 6.5 (consistency anchors). (i) Free field ($\lambda = 0$): $\langle \phi_0(f)\Omega_0, e^{-tH_0}\phi_0(f)\Omega_0 \rangle = \langle f, (2\mu)^{-1}e^{-t\mu}f \rangle \leq e^{-m_0 t} \langle f, (2\mu)^{-1}f \rangle$; the rate equals the gap m_0 , and Theorem 6.2 with $m_1 \uparrow m_0(1 - \varepsilon)$ (GJS Thm. 1.1.2 as $\lambda \rightarrow 0$) is consistent. (ii) `m3/m3.tex` Thm. 2.3 provides the converse direction (gap $\Rightarrow$ clustering) with rate $\gamma/2$; the present derivation (clustering $\Rightarrow$ gap) loses nothing in the rate, reflecting the general fact that Euclidean decay transfers to the spectrum without the factor-2 cost of the real-time filter argument. (iii) By `m3/m3.tex` Thm. 4.2 (the $\mathbb{Z}_2$ dichotomy), Theorem 6.2 implies that in the GJS weak-coupling region the $\mathbb{Z}_2$ symmetry is *unbroken* along every cofinal sequence — as expected, and a useful cross-check that the weak-coupling region is on the right side of the dichotomy.

7 Status

- **Proved here** (per fixed g , constants explicit): the two-time slab transfer (Lemma 3.1); $\hat{\psi}_T \rightarrow \Omega_g$ without isolation (Lemma 3.2); second-moment domains (Lemma 3.3); one-point limits (Lemma 3.4); the lsc two-point upper bound (Lemma 3.5); the decay estimate (Proposition 4.1); the density theorem (Theorem 5.2); the spectral lemma (Lemma 6.1); and Theorem 6.2.
- **Inputs:** printed GJS Thm. 1.1.7 (class-uniform as stated); Gate R1 for the auxiliary second moments; the conventions lock of `m2-audit.tex`. Gate R2 affects only constants.

- **Consequence:** together with `m2-audit.tex`, both hypotheses (G) and (M₂) of `lpp1.tex` are now derived from the same two printed source theorems, modulo the single bounded verification Gate R1. The weak-coupling closure of `weak-coupling-strategy.tex`, Thm. 7.1, is therefore conditional on Gate R1 alone.

References

- [1] J. Glimm, A. Jaffe and T. Spencer, *The Wightman axioms and particle structure in the $\mathcal{P}(\phi)_2$ quantum field model*, Ann. of Math. **100** (1974) 585–632.
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